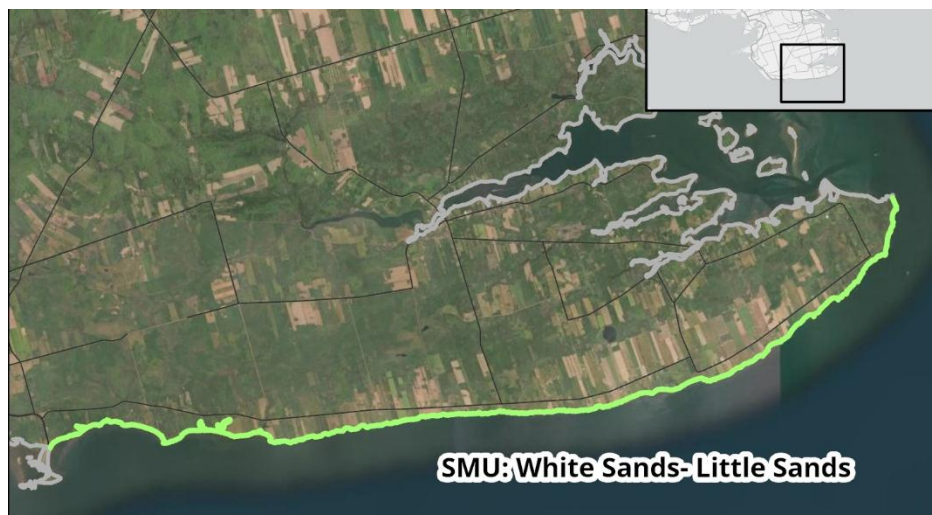


25. White Sands – Little Sands

Shoreline length: 38 km



Shore Types

White Sands – Little Sands is an exposed stretch of coast at the southeastern end of PEI with mostly cliffs and bluffs.

% Shore types within SMU

low plain	bluff	sloped armour	cliff	wetland	vertical struct.	dunes	sandspit
6	16	0	66	5	0	7	0



Bluff at Wood Islands Provincial Park



Cliff at Cape Bear Lighthouse and Marconi Station



Cliffs along community of White Sands

Governance, Infrastructure, and Economy

Approximately 15% of this area is located within the Rural Municipality of Belfast; the remainder is unincorporated. Key infrastructure includes roads accessing beaches as well as roads extending along the

shoreline: Panting Shore Road at the beach; Shore Road (Route 4), east of the Buddhist Institute at Little Sands; MacFarlane Road (White Sands, at the beach); and Black Brook Road (Beach Point). Coastal economic drivers are limited. Local and legacy land use constraints include numerous small lots and unpaved coastal roads.

Natural Assets and Heritage

Key features include the Northumberland Provincial Park (playground, campground), the Cape Bear Lighthouse, the Marconi Station, and Guernsey Cove Beach. The shoreline includes critical habitat for bird species including the threatened Bank.

Planning and Policy Framework

Provincial rules apply. Land use and development is managed under provincial development standards, which include horizontal coastal building setbacks based on edge of feature and a calculation of erosion over time, building setbacks for dunes, and a requirement for a coastal buffer for new subdivisions. There are no specific flood elevations

identified in regulations, but coastal hazard assessments are typically conducted.

Sediment transport

The sediment transport across the SMU is high, with eroding cliffs and bluffs feeding beaches typically from east to west.

Coastal Flood and Erosion Risk

The area has relatively low density of buildings and infrastructure within the coastal hazard zone. The small river mouths and coastal inlets to the west are vulnerable to increasing flood hazards. Erosion hazard is also highest along the exposed low plains.

The following flood risk indicator is based on the even with 1% annual exceedance probability (100-year return) with increasing amounts of sea level rise (SLR). The erosion risk indicator is based on potential future shoreline change for areas with historical data, not including salt marshes.

Estimated number of buildings and length of road at risk [per km shoreline].

	FLOOD RISK				EROSION RISK		
	Present	Short-term 0.3m SLR (2050)	Medium-term 0.6m SLR (2080)	Long-term 1.2m SLR (2130)	Short-term (~2050)	Medium-term (~2080)	Long-term (~2130)
Buildings	2 [<1/km]	3 [<1/km]	3 [<1/km]	6 [<1/km]	1 [<1/km]	9 [<1/km]	23 [1/km]
Roads	869 m [23 m/km]	957 m [25 m/km]	1051 m [27 m/km]	1231 m [32 m/km]	633 m [17 m/km]	1108 m [29 m/km]	1734 m [45 m/km]

Proposed SMP Strategies

No Active Intervention is proposed for the short-term, followed by Managed Realignment for the limited number of assets at risk.

	Short-term	Medium-term	Long-term
Sea level rise range	0 – 0.3 m	0.3 – 0.6 m	0.6 – 1.2 m
Potential timeframe	2030 - 2050	2050 - 2080	2080 - 2130
25.00 White Sands-Little Sands	NAI	MR	MR
Rationale	No Active Intervention is proposed for the short-term, followed by Managed Realignment. This will reduce risk to the limited number of vulnerable assets while preserving natural shorelines, including beaches and cliff habitat.		

The table above shows how the recommended shoreline management unit (SMU) or policy unit (PU) strategy evolves across three planning epochs (Short-, Medium-, and Long-term) under increasing sea-level rise and erosion projections.

Adaptation strategies are preliminary recommendations subject to further development with feedback from community and government. These strategies are not approvals, commitments or policy determinations.